

RayGlides 3W/4W EV Kit Cooling Controller - System Block Diagram

1. System Block Diagram (Mermaid)

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graph TD
    %% Power Section
    V_BAT[48V Vehicle Battery] -->|Fuse 2A / TVS / Filter| R_POL[Reverse Polarity & Surge Protection]
    R_POL -->|Raw Power 36V-72V| BUCK1[LM5017 12V Buck Regulator]
    BUCK1 -->|12V Power Rail| FAN_PWR[12V Fan Connector]
    BUCK1 -->|12V Rail| BUCK2[AP63205 5V Buck Regulator]
    BUCK2 -->|5V Rail| LDO[AP2112K 3.3V LDO Regulator]
    BUCK2 -->|5V Rail| SENS_PWR[LM35 Sensor Power VCC]
    LDO -->|3.3V Rail| ESP32[ESP32-S3-N16R8 MCU]

    %% Signal Section
    SENS[LM35 Temp Sensor] -->|Analog V_OUT 10mV/C| FILT[1k/100nF RC Filter + TVS]
    FILT -->|Clean Analog Signal| ADC_PIN[ESP32 GPIO 4 / ADC1_CH3]

    ESP32 -->|GPIO 18 / 5kHz PWM| GATE_DRV[Gate Drive 100R/10k]
    GATE_DRV -->|PWM Gate Drive| MOSFET[A03400A Low-Side Switch]
    MOSFET -->|Low-Side PWM Control| FAN_CONN[Fan Connector Pin 2 / Flyback Diode]

    %% Debug and Communication
    ESP32 <-->|GPIO 9/10| CAN_BUS[CAN Bus Transceiver]
    ESP32 <-->|GPIO 11/12/13| RS485[RS485 UART Transceiver]
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2. Power-Distribution Hierarchy

- 48V Nominal Input Stage**: Accepts unregulated voltage from the main lithium traction pack (range: 36V to 72V). Bypasses automotive load-dumps and switching ripple via a robust TVS-clamped passive pi-network filter.
- 12V Intermediate Power Stage**: Generated via TI's LM5017 100V-input synchronous buck regulator. This rail supplies the 12V DC brushless fan directly and serves as the input to the secondary 5V regulator.
- 5V Logic Power Stage**: Generated via Diodes Inc. AP63205 synchronous buck regulator from the 12V rail. Powering the LM35 sensor from 5V ensures linear operating headroom for the analog outputs.
- 3.3V Microcontroller Stage**: Generated via Diodes Inc. AP2112K-3.3 high-accuracy LDO regulator from the 5V rail. This isolates the sensitive analog-to-digital converter (ADC) references from any transient switching noise generated by the ESP32 module's RF operations.